

DEER LAKE, NL, Canada

WMO: 718090

Lat: 49.211N

Lon: 57.397W

Elev: 22

StdP: 101.06

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.6	-18.4	-25.8	0.4	-21.0	-22.3	0.5	-17.7	13.2	-4.4	11.5	-5.6	1.1	60	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	27.0	18.9	25.5	18.1	23.9	17.4	20.5	24.8	19.5	23.5	18.6	22.2	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.8	18.0	13.0	21.7	17.0	12.1	20.9	59.3	24.9	55.9	23.6	52.7	22.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.1	8.2	DB	-27.8	30.4	4.4	1.4	-31.0	31.4	-33.5	32.2	-36.0	33.0	-39.2	34.1	(4)
(5)			WB	-28.1	22.3	4.3	0.8	-31.2	22.8	-33.7	23.3	-36.1	23.7	-39.2	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-7.0	-7.6	-4.2	1.6	7.2	12.5	17.2	16.7	12.0	6.5	1.6	-3.4	(6)	
	DBStd	9.49	5.11	5.67	5.21	3.66	3.37	3.85	3.05	3.05	3.72	3.58	4.19	4.71	(7)	
	HDD10.0	2642	528	492	439	251	101	17	1	1	20	122	254	416	(8)	
	HDD18.3	5096	787	726	698	501	344	179	58	68	191	368	502	674	(9)	
	CDD10.0	632	0	0	0	1	14	91	223	207	81	12	2	0	(10)	
	CDD18.3	45	0	0	0	0	0	4	22	17	2	0	0	0	(11)	
	CDH23.3	439	0	0	0	0	6	62	216	139	17	1	0	0	(12)	
	CDH26.7	55	0	0	0	0	0	7	29	17	2	0	0	0	(13)	
(14) Wind	WSAvg	3.6	4.0	4.0	4.1	4.1	3.7	3.5	3.3	3.0	3.3	3.3	3.6	3.8	(14)	
(15) Precipitation	PrecAvg	1105	106	79	76	65	72	84	84	104	95	110	111	115	(15)	
	PrecMax	1298	162	210	134	111	105	160	209	167	192	193	174	200	(16)	
	PrecMin	811	53	24	13	25	37	25	21	50	32	49	58	54	(17)	
	PrecStd	126	32	43	31	27	21	38	44	31	36	38	31	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	8.3	8.9	15.8	23.7	27.6	29.1	28.6	25.4	19.9	16.7	9.4	(19)	
		MCWB	8.6	6.8	5.6	10.7	14.8	17.8	20.4	20.0	18.9	15.3	14.2	8.2	(20)	
	2%	DB	5.2	4.4	6.3	12.2	20.2	25.1	27.4	26.5	22.8	16.9	13.2	6.3	(21)	
		MCWB	4.2	3.2	3.7	7.8	12.5	16.5	19.2	19.2	17.4	13.6	11.5	5.2	(22)	
	5%	DB	2.3	2.1	4.3	9.6	17.4	22.9	25.9	24.9	20.6	14.5	10.2	4.2	(23)	
		MCWB	1.3	0.8	1.9	6.0	11.3	15.5	18.3	18.2	16.3	12.0	8.8	3.1	(24)	
	10%	DB	0.3	0.5	2.6	7.5	14.9	20.7	24.2	23.1	18.9	12.7	7.5	2.6	(25)	
		MCWB	-0.6	-0.6	0.8	4.2	10.1	14.8	17.8	17.3	15.5	10.5	6.3	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	7.1	6.7	11.6	15.9	20.1	22.2	21.8	20.3	16.5	14.5	8.4	(27)	
		MCDB	9.6	8.0	8.2	14.4	20.5	24.9	27.4	26.2	23.5	18.4	16.4	9.3	(28)	
	2%	WB	4.3	3.4	4.2	8.5	13.8	18.3	20.8	20.3	18.7	14.2	11.6	5.4	(29)	
		MCDB	5.1	4.2	5.8	11.2	18.2	22.3	25.0	24.7	21.5	16.3	13.1	6.2	(30)	
	5%	WB	1.3	1.0	2.3	6.3	12.1	16.9	19.7	19.3	17.2	12.4	9.1	3.2	(31)	
		MCDB	2.2	1.9	3.9	9.1	16.4	21.7	23.7	23.2	19.9	14.3	10.2	4.0	(32)	
	10%	WB	-0.6	-0.5	0.9	4.5	10.3	15.3	18.7	18.4	15.7	10.6	6.3	1.7	(33)	
		MCDB	0.3	0.4	2.4	7.1	14.7	19.7	22.6	21.9	18.4	12.5	7.4	2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.2	9.3	8.4	11.0	11.5	11.2	10.8	10.2	8.5	6.7	6.5	(35)	
		MCDBR	10.2	9.7	11.0	12.7	17.5	16.7	15.2	13.8	13.2	12.2	10.5	8.1	(36)	
	5% WB	MCWBR	9.9	8.9	8.9	8.9	10.9	9.3	8.2	7.6	8.4	9.2	9.3	7.9	(37)	
		MCWBR	10.2	9.2	9.5	11.5	15.0	13.5	12.0	11.6	11.2	11.1	10.3	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.358	0.341	0.388	0.385	0.388	0.427	0.403	0.377	0.339	0.317	0.304	(40)	
		taud	2.224	2.015	2.225	2.221	2.342	2.358	2.286	2.347	2.417	2.485	2.435	2.265	(41)	
	Ebn at Noon	Ebn at Noon	736	750	862	856	875	872	830	834	821	798	729	674	(42)	
		Edh at Noon	76	120	115	129	119	118	126	113	96	75	63	65	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.85	1.57	2.80	3.97	4.82	5.08	5.09	4.26	3.19	1.95	0.99	0.61	(44)	
	RadStd	RadStd	0.08	0.23	0.25	0.46	0.34	0.55	0.40	0.29	0.26	0.16	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.63	N/A	+0.54	+0.71	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	+1.24	N/A	+0.38	+0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page